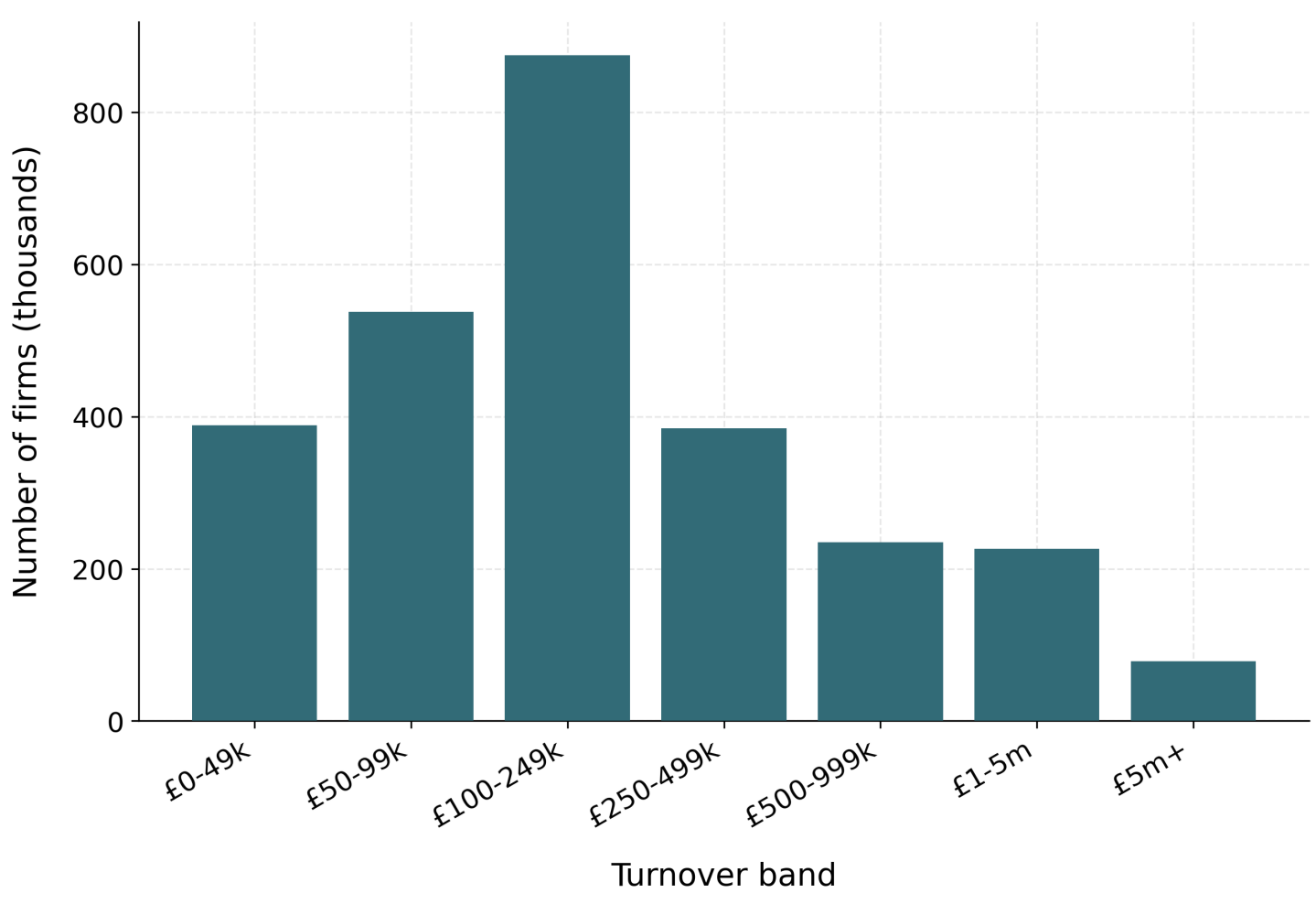




The problem: VAT as a tax *notch*

The UK VAT registration threshold is a **notch**, not a kink. Once taxable turnover crosses T^* , **20% VAT falls due on the whole turnover**, not just the increment — a discrete liability jump of $\tau T^* = \text{£17,000}$ at $T^* = \text{£85,000}$. Threshold **£85,000** (2023–24 data year), raised to **£90,000** in April 2024 — among the **highest in the OECD** — and it sits in a *dense* part of the firm-size distribution. Firms respond by **locating just below** it; the cost of crossing is sharpest for *consumer-facing* firms with little reclaimable input VAT, so the notch distorts both the size distribution and firms' growth at the margin.

All UK firms by turnover band — ONS, 2023–24



Three reform families, one basis

What we cost.

Costed on a *common static basis*:

- **Level** — move the threshold up or down.
- **Shape** — a graduated taper that phases VAT in.
- **Rate** — a reduced-rate band above the threshold.

Contribution

Headline: only the graduated taper removes the notch's distortion; the level move is the dearest and least efficient.

An open, reproducible firm-level microsimulation that:

1. costs **level + shape + rate** reforms on one static basis;
2. characterises the notch's distortion via its exact, **elasticity-free dominated region**;
3. shows via a **placebo** that synthetic bunching is mechanical;
4. adds a conditional **behavioural layer**.

Data methodology

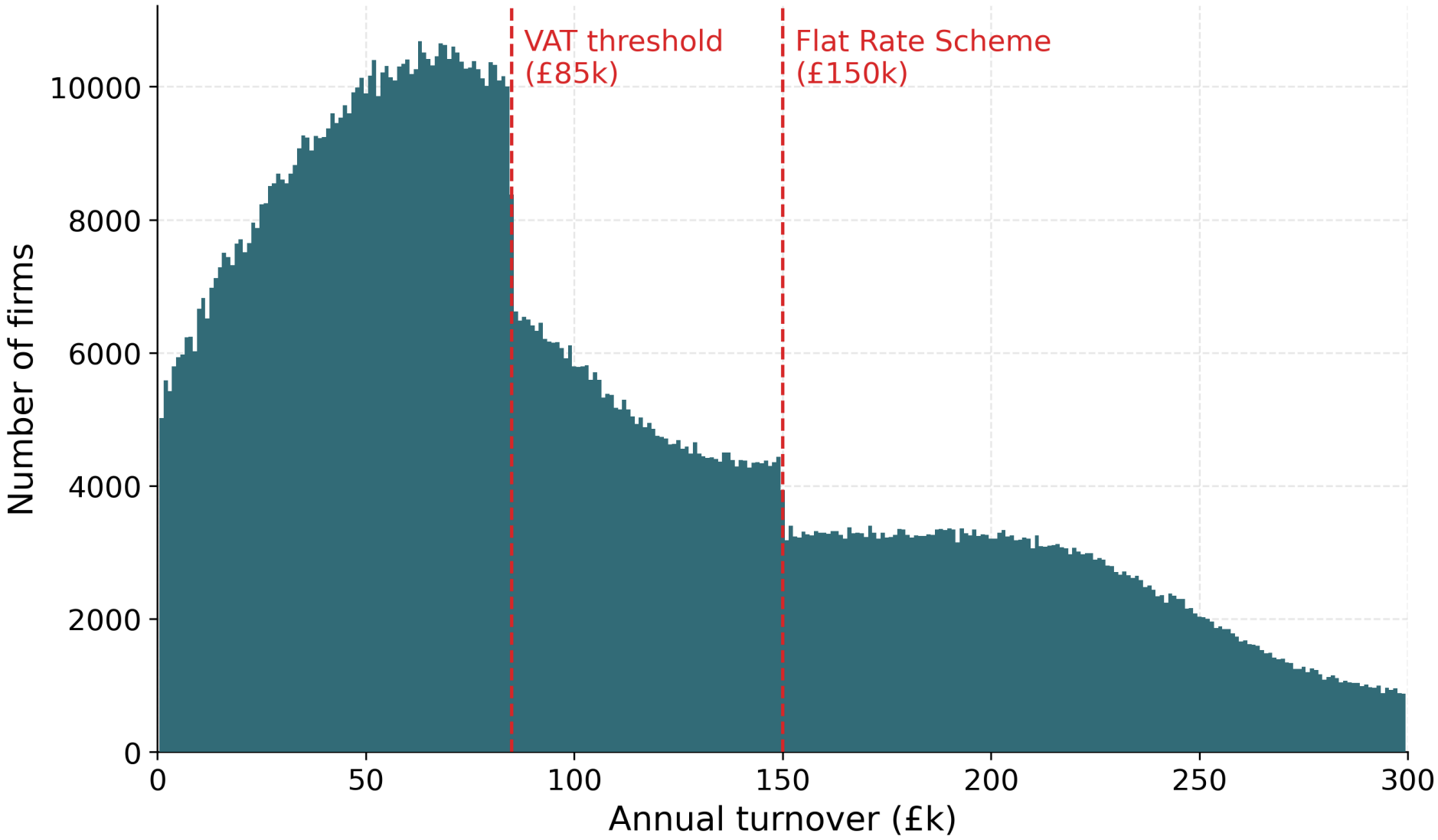
How the synthetic data are built.

Draw synthetic firms by sector $\times$ turnover band; give each an input share $x_i = \rho_i y_i$ so value added $v_i = y_i - x_i$ is the net VAT base. Firm weights $w_i = e^{\theta_i}$ are fit by gradient descent to HMRC + ONS aggregates. Crucially, the threshold is a *generator parameter*: every counterfactual — a higher cutoff, a taper, a reduced rate — **regenerates** a fresh, internally consistent population, and the whole file is **released openly**.

Synthetic firm data

→ *The engine behind every costing.*

$\approx 2.94\text{m}$ firm rows weighted to $\approx 2.5\text{m}$ UK firms, calibrated to HMRC + ONS aggregates (turnover-band counts, sector, employment, net VAT liability by band) at $\approx 90\%$ **accuracy**. Calibrated to the aggregates, so agreement is an **internal-consistency check**, not external validation.



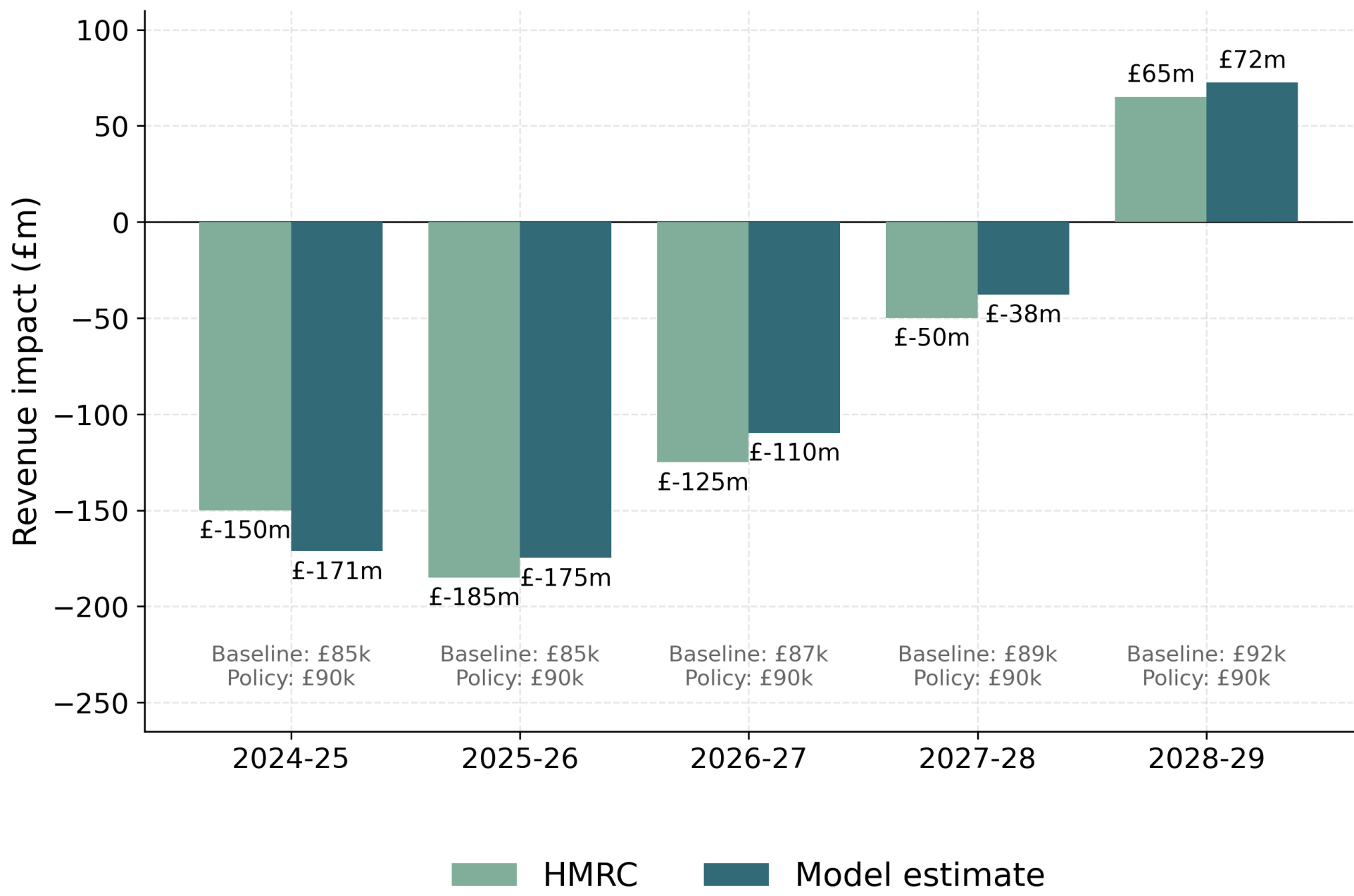
Static costing

Pricing a reform on this data.

$$R(T) = \sum_i w_i v_i \mathbb{1}\{y_i \geq T\}$$

Revenue moves only through firms whose registration status flips.

Validation vs HMRC: the £85k→£90k April-2024 increase costs **–£175m** in 2025–26 vs HMRC's published **–£185m** — agreement within $\sim \text{£10m}$.



Threshold sweep (2025–26)

Scaling the level reform across thresholds.

From a £90k base (**£203.8bn**), the static cost is roughly **linear**, $\approx \text{£150} - \text{£220m}$ per £5k step — e.g. **–£374m** to raise to £100k, **+£665m** to lower to £70k.

Reform menu

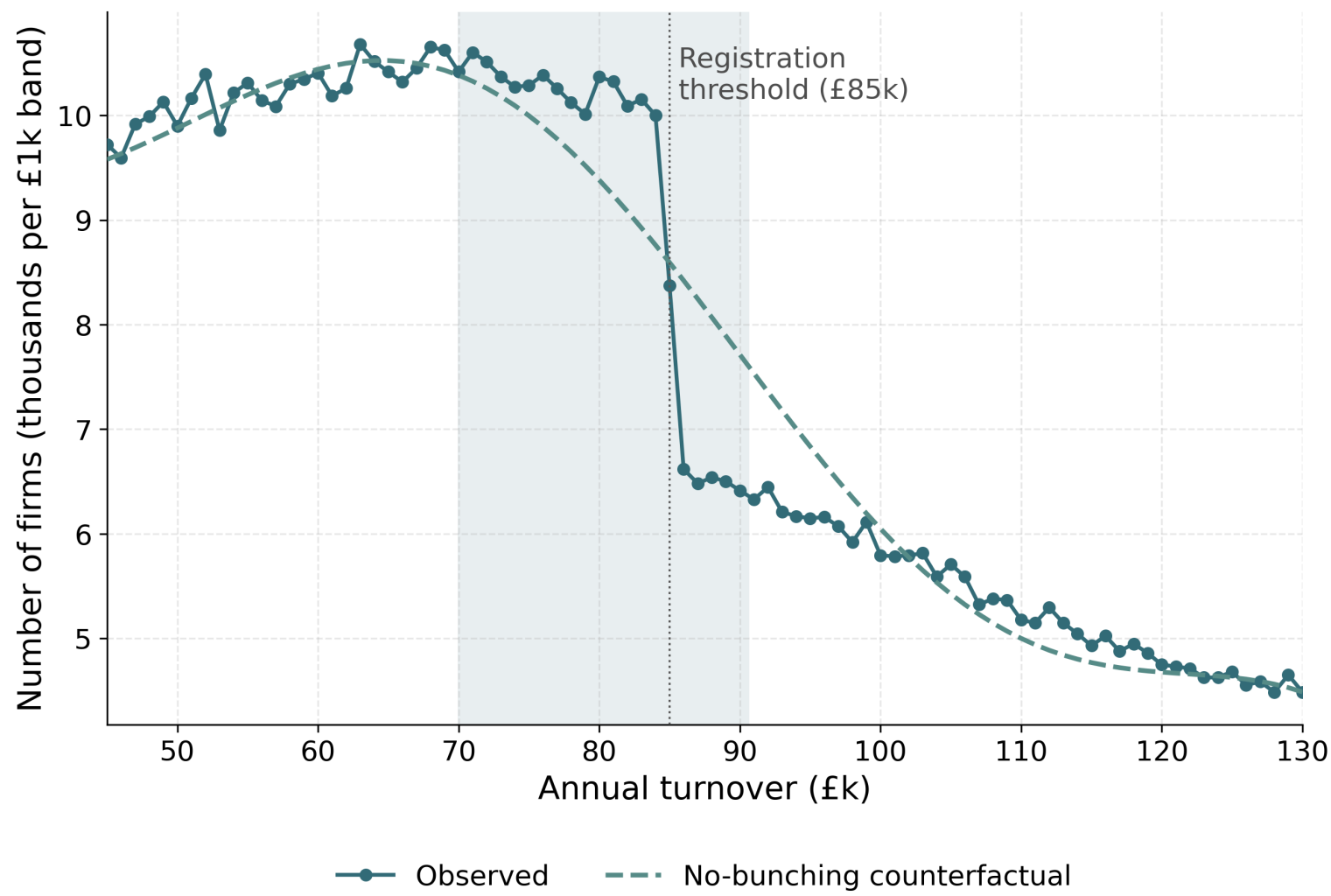
→ *Cost and distortion on a common £85k / £183.6bn base.*

Reform	Type	Static	Dominated region
Raise to £100k	Level	–£508m	relocated
Taper £85–105k	Shape	–£336m	removed
Reduced 10%	Rate	–£343m	split in two
Reduced 15%	Rate	–£171m	split in two

Bunching is mechanical

A transparency check on the synthetic data.

Estimator: $b = 0.060$, excess mass $E = 8,712$ firms. A **placebo** collapses E to **0–99** when the step is removed from the calibration target $\Rightarrow$ bunching is inherited from coarse HMRC band targets, *not* firm behaviour. Behavioural bunching is therefore **not identified** on the synthetic data; the administrative estimate of Liu et al. (2021) remains the behavioural fact.



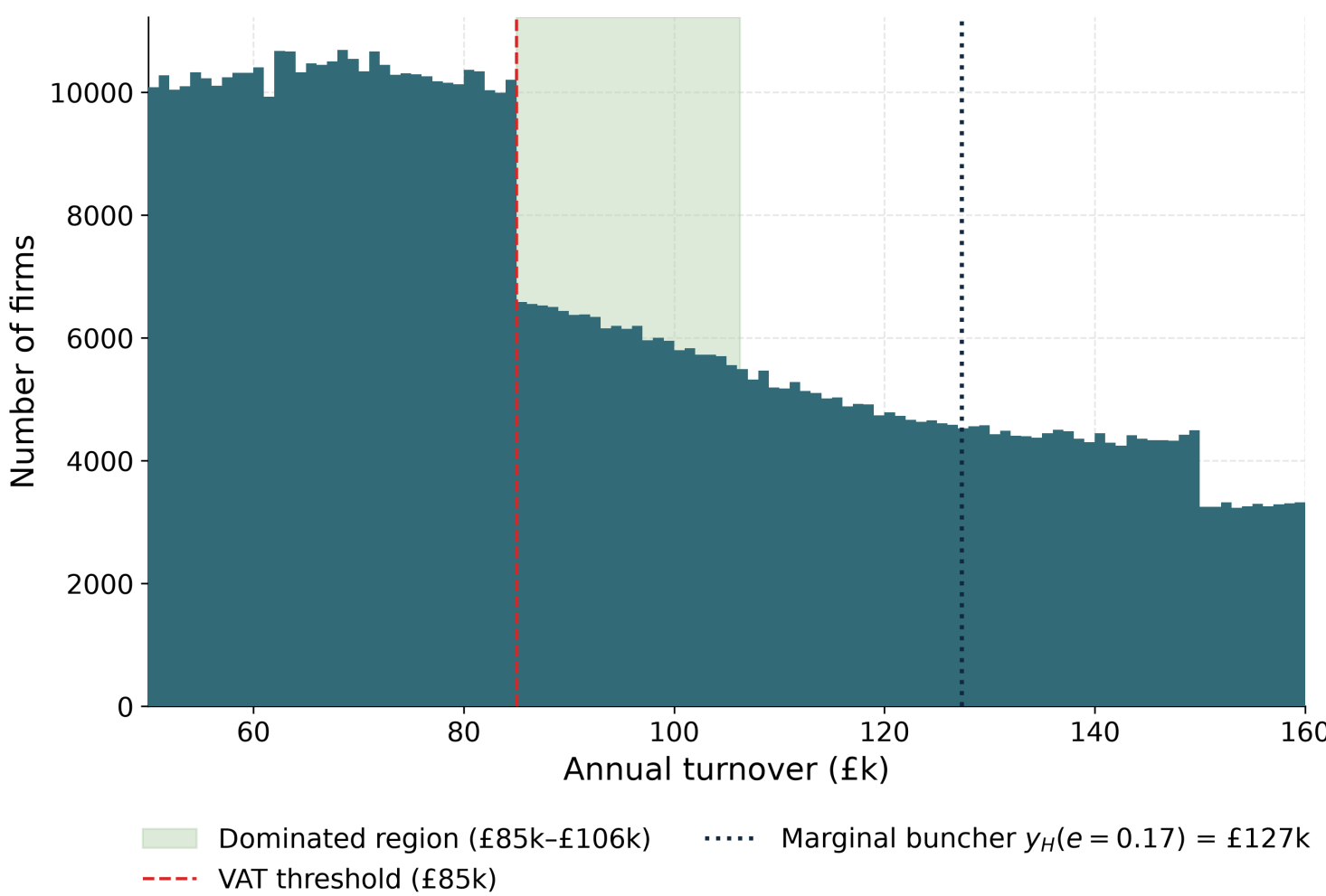
The dominated region

Beyond revenue: measuring the notch's distortion.

Profit jumps *down* by $\tau T^* = \text{£17,000}$ at the threshold. The **dominated region** — turnover where no firm locates because it earns less than at T^* — has exact width

$$a = T^* \frac{\tau}{1-\tau} = \text{£21,250}$$

(£85,000–£106,250), spanning $\approx 137,000$ weighted firms. **Exact and elasticity-free** (conditional on the turnover-tax notch).



How reforms change it

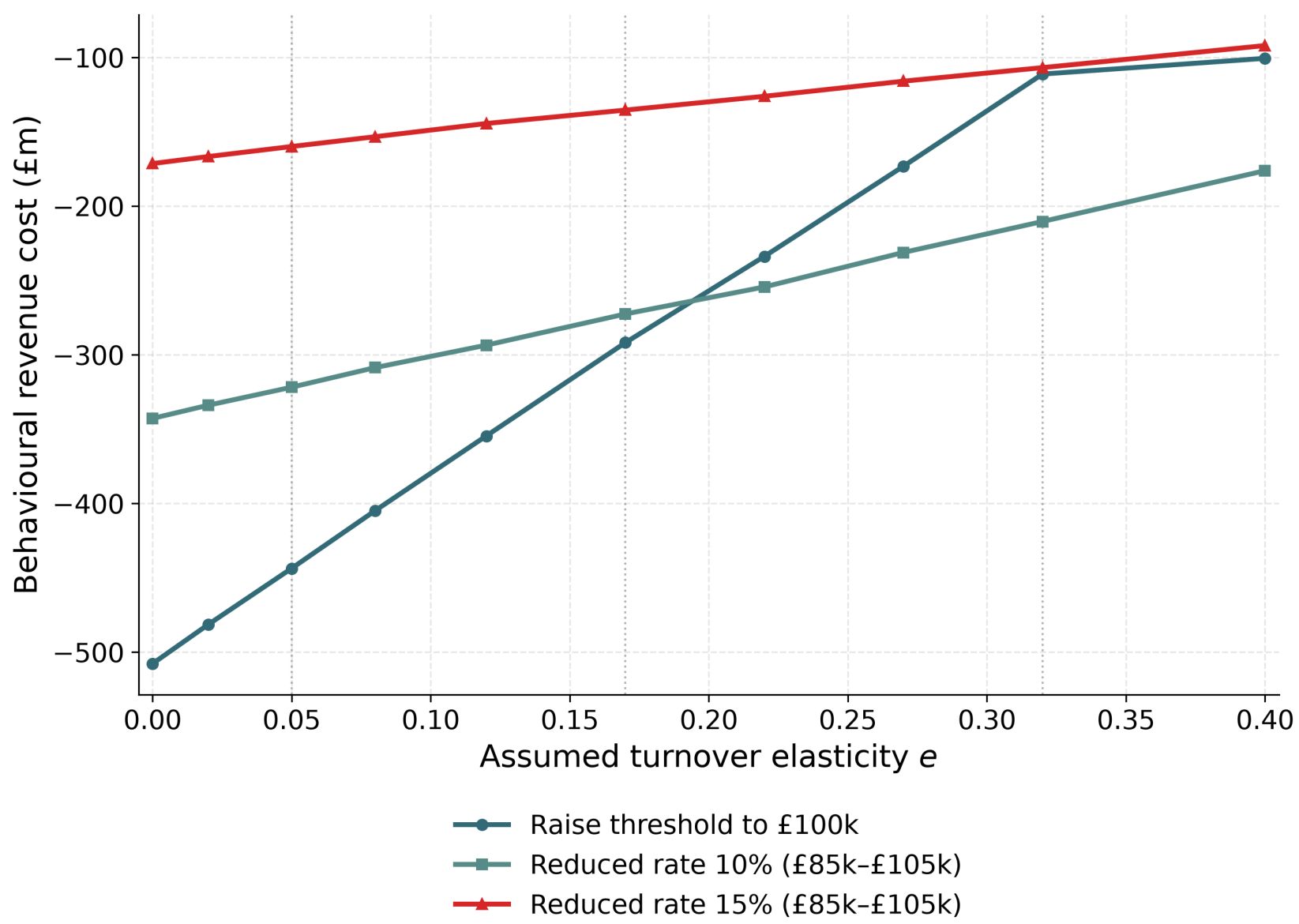
→ *Which reforms actually shrink this band?*

- Raising the threshold **relocates** the £21,250 band unchanged.
- A banded reduced rate does **not shrink** it. It lowers the *largest single* notch (£17,000→**£12,750** at 15%), but reverting to 20% at the band top adds a *secondary* notch $a' = T_1 \frac{\tau}{1-\tau}$ of £5,250, so total width is **£21,562 (15%) / £22,569 (10%)** — it **splits** the band, not shrinks it.
- **Only the graduated taper removes the distortion** entirely.

Behavioural layer

Layering an assumed response on the static costs.

Conditional on an **assumed** turnover elasticity e (*not* identified on synthetic data); sweep $e \in \{0.05, 0.17, 0.32\}$. At the midpoint $e = 0.17$, each static cost falls as firms broaden the base: raising to £100k from £508m to **£292m**; the 10% band £343m to **£273m**; the 15% band £171m to **£135m**. A larger e makes every reform **cheaper**. The taper is *excluded* — its base-broadening and marginal-rate channels cannot be netted in the iso-elastic cost. Only $e = 0.05$ has an external anchor, so this is a **sensitivity range**, not a forecast.



Conclusions

What this paper is: an open, reproducible microsimulation that models the UK VAT threshold as a *turnover-tax notch* on synthetic firm data, and prices level, shape and rate reforms three ways — statically, by the exact distortion each removes, and behaviourally.

1. Prices level + shape + rate on one basis; reproduces HMRC's anchor costing.
2. On the dominated-region metric, **only the taper removes the distortion** — the level move is the dearest and least efficient.
3. Synthetic bunching is **mechanical** (a transparency lesson) — no elasticity is read off it.
4. Behavioural offsets are conditional on an assumed e ; the static cost and dominated region are the **elasticity-free anchors**.